


Exam 2 on Friday!

- Bring a calculator!
- around 10-12 questions, some with a few parts
 - a couple short answer questions at start
 - then the rest are step-by-step data analysis for a given data set (very similar format to Exam 1)

Equations to know:

• R^2 and R^2_{adj}

• VIF $\leftrightarrow R^2$

• t statistic $\leftarrow$

$$t = \frac{\hat{\beta}_j - (\text{value under } H_0)}{SE \hat{\beta}_j}$$

• F statistic $\leftarrow$

$$\frac{(SSE_{\text{reduced}} - SSE_{\text{full}}) / (P_{\text{full}} - P_{\text{reduced}})}{\frac{1}{n - P_{\text{full}}} SSE_{\text{full}}}$$

• Fill in ANOVA table

• SSE , SST_{total} , SS_{Model}

$$\sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad \downarrow \quad \sum_{i=1}^n (y_i - \bar{y})^2 \quad \rightarrow \quad \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

Adding parameters + SSE, R^2 , R^2_{adj} :

$$R^2 = 1 - \frac{SSE}{SSTotal}$$

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

as we add variables (parameters) to the model (make it larger / more flexible): $SSE \downarrow$ (even if only by a very small amount)

$$\Rightarrow R^2 = 1 - \frac{SSE}{SSTotal} \quad \uparrow$$

(R^2 can never decrease when we add more terms, b/c SSE can never increase when we add more terms)

$\Rightarrow$ Problem for comparing models w/ different #s of explanatory variables

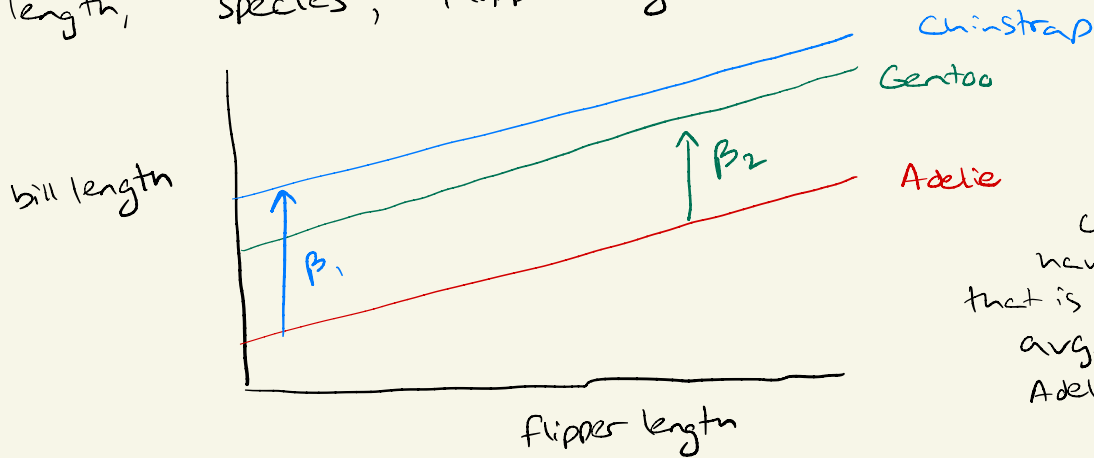
Solution: instead of comparing R^2 , compare R^2_{adj} between models

$$R^2_{adj} = 1 - \frac{SSE / (n - p)}{SSTotal / (n - 1)} \quad \leftarrow \text{penalty for \# parameters in model}$$

if we add more terms, $SSE \downarrow$, but $p \uparrow \Rightarrow n - p \downarrow$
 $\Rightarrow SSE / (n - p)$ could either increase or decrease
 $\Rightarrow R^2_{adj}$ can either increase or decrease

Categorical explanatory variables

bill length, species, flipper length



Chinstrap

Gentoo

Adeline

β_1 : for a fixed value of flipper length, chinstrap penguins have an avg. bill length that is different from avg. bill length of Adeline penguins by β_1

$$\text{bill length} = \beta_0 + \beta_1 \text{Is Chinstrap} + \beta_2 \text{Is Gentoo} + \beta_3 \text{Flipper Length} + \varepsilon$$

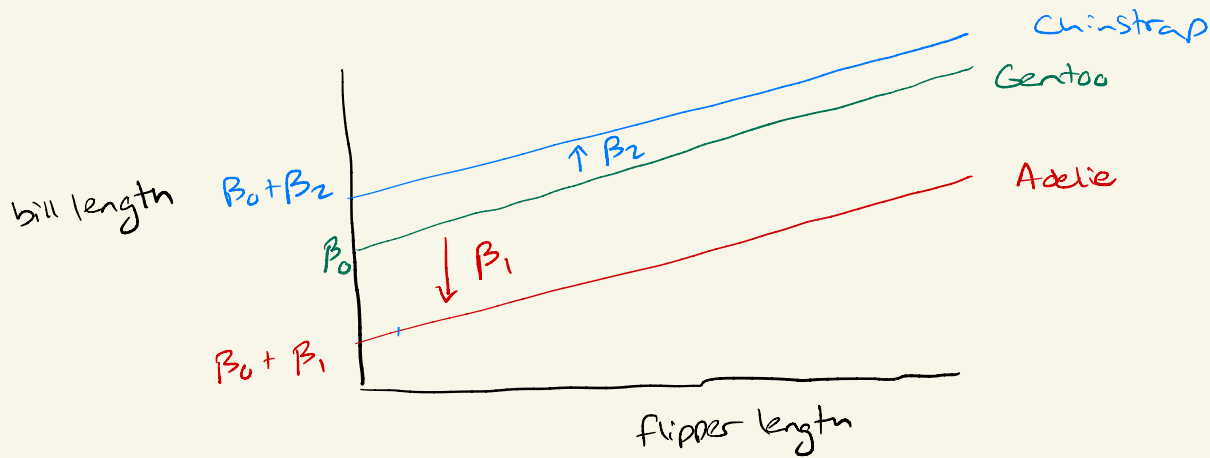
$$\text{Is Chinstrap} = \begin{cases} 0 & \text{not chinstrap} \\ 1 & \text{Chinstrap} \end{cases}$$

$$\text{Is Gentoo} = \begin{cases} 0 & \text{not Gentoo} \\ 1 & \text{Gentoo} \end{cases}$$

Adeline: $\text{Is Chinstrap} = \text{Is Gentoo} = 0$
 $\Rightarrow \beta_0 + \beta_3 \text{Flipper length} + \varepsilon$
 (intercept = β_0)

Chinstrap: $\text{Is Chinstrap} = 1, \text{Is Gentoo} = 0$
 $\Rightarrow \beta_0 + \beta_1 + \beta_3 \text{Flipper length} + \varepsilon$
 (intercept = $\beta_0 + \beta_1$)

β_3 : holding species fixed, a one-unit increase in flipper length is associated with a change of β_3 in bill length on average



(in this picture,
 $\beta_1 < 0$
 $\beta_2 > 0$)

make gentoo reference group:

$$\text{bill length} = \beta_0 + \beta_1 \text{IsAdelle} + \beta_2 \text{IsChinstrap} + \beta_3 \text{Flipper length} + \epsilon$$

Currently : assuming that relationship between flipper length & bill length is same for each species
 (slope β_3 for all 3 lines)

To allow different slopes: need interaction

e.g.

$$\text{bill length} = \beta_0 + \beta_1 \text{IsAdelle} + \beta_2 \text{IsChinstrap} + \beta_3 \text{Flipper length} + \beta_4 \text{IsAdelle} \cdot \text{Flipper length} + \beta_5 \text{IsChinstrap} \cdot \text{Flipper length} + \epsilon$$

ANOVA table

(a single categorical explanatory variable)

$$\text{bill length} = \beta_0 + \beta_1 \text{lsChinstrap} + \beta_2 \text{lsGentoo} + \varepsilon$$

want to test $H_0: \beta_1, \beta_2 = 0$ (no relationship between species & bill length)

H_A : at least one of $\beta_1, \beta_2 \neq 0$

(some relationship between species & bill length, i.e. some difference in avg. bill between the three species)

F statistic:

$$\frac{MS_{\text{Model}}}{MSE}$$

$$= \frac{SS_{\text{Model}} / (p-1)}{SSE / (n-p)}$$

$p = \# \text{ parameters in model}$

$n = \# \text{ obs}$

what we want to calculate in ANOVA table

$$= \frac{(SS_{\text{Total}} - SSE_{\text{full}}) / (p-1)}{SSE_{\text{full}} / (n-p)}$$

(H_A)

full: $\beta_0 + \beta_1 \text{lsChinstrap} + \beta_2 \text{lsGentoo} + \varepsilon$

(H_0)

reduced: $\beta_0 + \varepsilon$

$$SS_{\text{Total}} = SSE_{\text{reduced}}$$

$\# \text{ params in reduced} = 1 (\beta_0)$

$$\Rightarrow F = \frac{(SSE_{\text{reduced}} - SSE_{\text{full}}) / (p_{\text{full}} - p_{\text{reduced}})}{SSE_{\text{full}} / (n - p_{\text{full}})}$$

Table

	df	SS	MS	F
Model	$p-1$	SS_{Model}	MS_{Model}	MS_{Model} / MS_E
Residuals	$n-p$	SS_E	MS_E	
Total	$n-1$	SS_{Total}		

 $n = \# \text{ obs}$ $p = \# \text{ params}$
(full model)

$$SS_{\text{Total}} = SS_{\text{Model}} + SS_E$$

$$MS_{\text{Model}} = \frac{SS_{\text{Model}}}{p-1}$$

$$MS_E = \frac{SS_E}{n-p}$$

- ① Start by filling in df (from description of data + full model (# of β s in model))
- ② I will give you some combo of SS_{Model} , SS_E , MS_{Model} , MS_E , F (two pieces)
- ③ Fill in rest